GGG-D-671d
April 10, 1964

SUPERSEDING
Int. Fed. Spec. GGG-D-00671c (Navy-Ships)
August 29, 1961 and
Fed. Spec. GGG-D-671b,
September 19, 1957

FEDERAL SPECIFICATION

DRILL; HAND, AND PUSH

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hollow- and solid-handled hand and push drills with single- or two-speed drives and ratcheting or non-ratcheting features as specified hereinafter.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The drills shall be of the following types, classes, and styles as specified (see 6.2):

Type I.—Hand, hollow wooden handle.
  Class 1.—Single speed.
    Style A.—With ratchet.
    Style B.—Without ratchet.
  Class 2.—Two-speed, without ratchet.
Type II.—Hand, solid wooden handle.
  Class 1.—Single speed, without ratchet.
  Class 2.—Two speed, without ratchet.

Type III.—Hand, all angle adjustable hollow handle.
Type IV.—Push automatic.
  Class 1.—Metal hollow handle.
  Class 2.—Plastic hollow handle.
  Class 3.—Wood hollow handle.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Standards:
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Hand-

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books and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:
MIL-H-15424—Hand Tools; Packaging of.

Military Standard:
MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of drills otherwise conforming to this specification.

3.2 Design. The drills shall be of the latest design, convenient to operate and entirely adequate for the purpose intended (see 1.1).

3.3 Material. The material of which the drills are constructed shall be of a quality to produce drills conforming to the requirements as hereinafter specified. The strength and other physical properties shall be adequate throughout to withstand, without injury, the maximum strain to which the drill may be subjected under ordinary service conditions.

3.4 Finish. All drills shall have a smooth even surface, thoroughly cleaned and substantially free from scale, pits, nodules, burrs, cracks, or other detrimental defects. Surfaces not usually ground or buffed shall have a surface finish conforming to good manufacturing practice. Ends and heads of pins and rivets shall have no projecting burrs.

3.4.1 *Coating.* The housing or cast portion of the frame and the nonmachined surfaces of the driving gear shall be enameled or coated with an equally suitable finish. Unless otherwise specified (see 6.2) the chuck crank arm, and ferrules shall be plated with nickel or chromium. Other exposed bare metal surfaces shall be plated with nickel or chromium or have a coating of clear lacquer. When nickel and chromium are restricted by government order, any one of the following coatings will be acceptable for protection from rust and corrosion:

(*a*) cadmium plate
(*b*) zinc plate
(*c*) hot phosphating
(*d*) Oxide coating

3.5 Repair parts. The successful bidders supplying drills covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.6 Threads. All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement.

3.7 Marking for identification. Drills shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.8 Type I, hand, hollow wooden handle. Type 1 drills shall consist essentially of a frame, spindle, chuck, crank, wooden handles, side grip or knob, and driving, idler, and pinion gears. The drills shall have an overall length of not less than 11-1/2 inches

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or more than 16 inches with a maximum weight of 5 pounds.

3.8.1 *Frame.* Unless otherwise specified (see 6.2), the frame shall be of the box or "H" design (see figures 2 through 6) and of malleable iron, steel or other metal suitable for the purpose, and of strong and durable construction. All bearing surfaces shall have a smooth machined finish and shall be free from any imperfections which may affect the durability or serviceability of the drill.

3.8.2 *Spindle.* The spindle shall be of high grade steel, smoothly finished and round within 0.001 inch. The total clearance between the spindle (diameter) and its bearing surface or surfaces shall be not more than 0.005 of an inch throughout the entire length of the bearing. A ball type bearing or a single ball shall be provided to absorb the axial thrust, and minimize spindle binding. The spindle and gears shall be designed and assembled so that the end play in the spindle will be not more than 0.010 of an inch. The spindle and its component parts shall be assembled in a manner to preclude the likelihood of their becoming loose under normal service conditions. The lower portion of the spindle shall have a fine series thread to properly accommodate the threaded hole of the chuck. These threads shall be concentric with the spindle within 0.005 inch total indicator reading throughout the entire length of thread.

3.8.3 *Chuck.* The chuck shall be steel with three hardened jaws capable of accurately and securely gripping the shanks of drill bits by hand tightening only. The jaws shall show a Rockwell hardness value of not less than 53 nor more than 62 on the C scale, or the jaws may be casehardened to show a hardness value of not less than 87 nor more than 91 on the Rockwell 15 N scale (superficial). Each chuck jaw in drills of 1/4-inch-capacity shall have a flat gripping surface .015 inch to .025 inch wide, and each chuck jaw in drills 3/8-inch capacity shall have a flat gripping surface .020 inch to .035 inch wide. The chuck shall be threaded on the spindle and designed to provide a means for opening and closing the chuck jaws by hand. Chuck jaw gripping surfaces shall be actuated by a hardened follower and shall be capable of holding all drills or bits within the rated capacity of the drill and within .030 of an inch total run out (see 4.3.1). Springs or other suitable means shall be provided to hold the jaws firmly against the tapered bearing surface on the inside diameter of the sleeve. The portion of the interior surface of the church against which the jaws bear shall be sufficiently hard to prevent the jaws from cutting into the sleeve or body. The outside diameter of the sleeve shall be knurled in a manner to provide a good handgrip. The chuck shall be threaded internally with a fine thread for attaching the chuck to the spindle.

3.8.4 *Crank arm.* The crank arm shall be of steel. The arm of the crank, measured from the center of the pivot to the center of the handle, shall be not less than 3 nor more than 4 inches for the 3/8-inch capacity drills and not less than 2 nor more than 3-1/2 inches for the 1/4-inch capacity drills.

3.8.5 *Wooden handles and knobs.* All wooden handles and knobs shall be of hardwood, properly sized and formed and sanded to a smooth finish. Handles and knobs may be color stained and shall be coated with varnish or lacquer to minimize water absorption.

3.8.5.1 *End gripping handle.* The end gripping handle shall be firmly attached to the frame and fixed to a steel ferrule provided for mounting and securing this handle. The end gripping handle shall be hollowed out to at least 1/2-inch diameter and 2-1/4-inches deep forming a magazine suitable for holding straight shank wood-boring drills. Means shall be provided for access to the drill magazine and be part of the handle.

3.8.5.2 *Cranking knob.* The cranking knob

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shall be provided with a steel ferrule and shall be mounted on, and free to rotate around the steel pin affixed to the end of the cranking arm.

3.8.5.3 *Drill point set.* Eight straight-shank, wood-boring drill points ranging in diameters from 1/16 to 11/64 inch inclusive, in increments of 1/64 inch, shall be furnished in the hollow-end gripping handle. These drill points shall be made from a good quality steel having a 0.70 to 0.80 percent carbon content and a reading on the Rockwell "C" scale of from 48 to 54. The drill points shall have two straight flutes at least 3/4 inch in length and an overall length of at least 1-3/4 inches. Drill points shall be similar to figure 1.

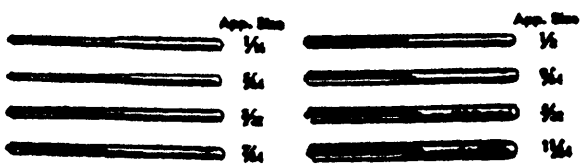


FIGURE 1.—Drill point set.

3.8.6 *Driving, idler, and pinion gears.* All gears shall have machine-cut beveled teeth free from any imperfections, burrs, or defects of any nature which may affect their serviceability. The face width of the gear teeth shall be not less than 23/64 inch (minimum) for the 3/8 inch capacity drills and 15/64 inch (minimum) for the 1/4-inch capacity drills with a maximum backlash of .020 inch. The driving gear shall be of malleable or gray iron with means provided for firmly attaching the crank. The idler gear and driven gear (pinion) shall be of steel. The gears shall run freely when assembled with means provided to counteract the outward thrust of the driving gear.

3.8.7 *Drill capacity.* The hand drills shall be furnished with a drill capacity of either 0 up to and including 1/4 inch, or 0 up to and including 3/8 inch, as specified (see 6.2).

3.8.8 *Class 1, single speed.* Class 1 drills shall have a gear ratio between the driving gear and driven gear (pinion) of not less than 3 to 1 for 1/4-inch capacity drills and 4 to 1 for 3/8-inch capacity drills.

3.8.8.1 *Style A, with ratchet.* Style A drills shall be furnished with a ratchet mechanism.

3.8.8.1.1 *Ratchet mechanism.* The ratchet mechanism shall be quick-acting and have a manually-operated knurled button or equally suitable means for permitting the selection of the directional rotation. The action of the chuck for the positions listed hereinafter shall be as follows:

(*a*) When the driving gear is positioned for right-hand rotation, the chuck shall rotate in a right-hand direction. No movement of the chuck shall take place with left-hand rotation of the driving gear.

(*b*) When the driving gear is positioned for left-hand rotation, the chuck shall rotate in a left-hand direction. No movement of the chuck shall take place with right-hand rotation of the driving gear.

(c) When the driving gear is positioned for either right- or left-hand rotation, the chuck shall rotate continuously in a right-hand direction.

(*d*) The ratchet mechanism shall provide for a neutral position for permitting a direct rotational drive of the chuck with either the right- or left-hand rotation of the driving gear.

(*e*) The ratchet mechanism shall provide a locking device for locking the driving gear and spindle.

3.8.8.1.2 Type I, class 1, style A drills shall be similar to figure 2.

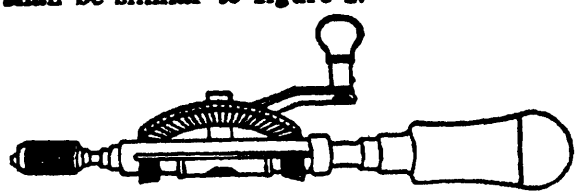


FIGURE 2.—Type I, class 1, style A, hand, hollow, wooden handle, single speed, with ratchet.

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3.8.8.2 *Style B, without ratchet.* Style B drills shall be furnished without a ratchet mechanism and shall be similar to figure 3.

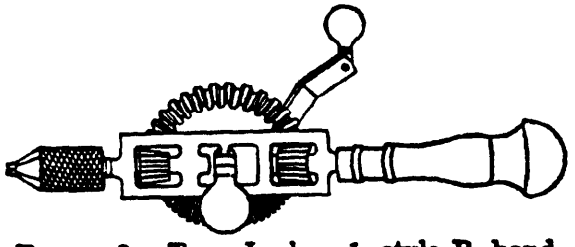


FIGURE 3.—Type I, class 1, style B, hand, hollow, wooden handle, single speed, without ratchet.

3.8.8.3 *Detachable sidegrips.* Type I, class 1, style B drills shall have detachable sidegrips which shall be made in accordance with 3.8.5, and provided with a steel ferrule. The sidegrip shall be located on the side of the frame opposite the driving gear and mounted on, and securely fixed to a steel shaft. This steel shaft shall be at least 3/16 inch in diameter and 1-1/4 inches long and shall have not less than 3/16 inch of threads projecting from the sidegrip. These shaft threads shall be suitable for mating with the threads in the hole provided in the frame.

3.8.9 *Class 2, two-speed without ratchet.* Class 2 drills shall have speed-changing mechanism and a gear ratio between the driving gear and driven gear (pinion) of not less than 0.75 to 1 for low speed and not less than 8 to 1 for high speed. Class 2 hand drills shall be furnished without a ratchet mechanism and shall be similar to figure 4.

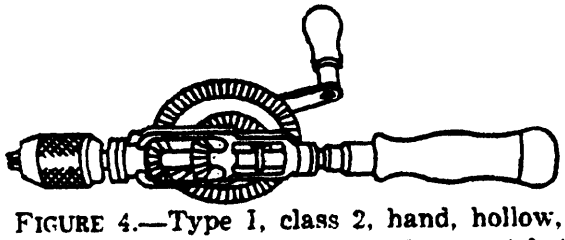


FIGURE 4.—Type I, class 2, hand, hollow, wooden handle, two speed, without ratchet.

3.8.10 *Speed-changing mechanism.* The speed-changing mechanism for the two-speed drills shall provide for a direct drive of the chuck in both the low- and high-speed operations. A manually-operated knurled button or equally suitable means shall be conveniently located for permitting the selection of either low or high speed.

3.9 **Type II, hand, solid wooden handle.** Type II drills shall conform to the requirements of 3.8 through 3.8.8 except the large or main handle shall be solid with no provision made in the handle for holding wood-boring drills.

3.9.1 *Class 1, single speed, without ratchet.* Class 1 hand drills shall be in accordance with 3.8.8 and 3.8.8.3, shall be furnished without a ratchet mechanism and shall be similar to figure 5.

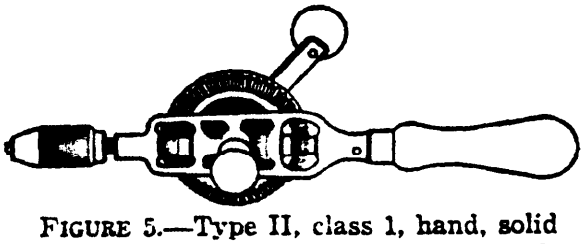


FIGURE 5.—Type II, class 1, hand, solid wooden handle, single speed, without ratchet.

3.9.2 *Class 2, two-speed, without ratchet.* Class 2 drills shall be in accordance with 3.8.10, shall be furnished without a ratchet mechanism, and shall be similar to figure 6.

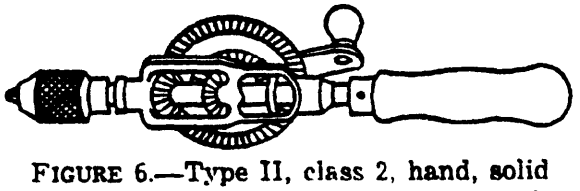


FIGURE 6.—Type II, class 2, hand, solid wooden handle, two speed, without ratchet.

3.10 **Type III, hand, all angle, adjustable hollow handle.** Type III drills shall consist essentially of a frame, chuck turret head, hollow handle, driving gear, crank arm, driven gear (pinion), spindle, and chuck. The drills shall be of high grade material either forged, sand- or die-cast, free from all defects and of a strong and durable construction in order to withstand the test requirements of 4.3. Drills constructed of sand- or die-cast material shall withstand additional test requirements of 4.3.3. The gear ratio between the driving gear and the driven gear (pinion) shall be not less than

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2.7 to 1 nor more than 3.5 to 1. The hand drills shall have an overall length of 10-3/4 inches plus or minus 1/4 inch, with a minimum weight of 1-1/4 pounds.

3.10.1 *Frame.* The frame shall be of metal suitable for the purpose intended (see 1.1). All bearing surfaces shall have a smooth, machine finish and shall be free from any imperfections which would interfere with their serviceability.

3.10.2 *Chuck turret head.* The chuck turret head shall be of metal and shall be provided with a minimum of six divisions or positions of offset equally divided about the centerline of the drill body capable of totaling 270 degrees of travel.

3.10.2.1 *Lock or catch.* A lock or catch of suitable design shall be provided to securely hold the chuck turret head in any position desired. The lock or catch shall be located in such a manner that it causes no obstruction during operation of the hand drill and shall be easily operated.

3.10.3 *Hollow handle.* The hollow handle shall be of metal. The pressure end of the handle shall be hollow and suitable for holding at least eight straight-shank wood-boring drills of a size within the capacity of the drill. A removable screwcap or other suitable means shall be provided for access to the drill magazine and be a part of the hollow handle. The handle shall be free to move through an included angle of 180 degrees in a plane perpendicular to the plane of movement of the chuck turret head. Tooth serrations or other suitable means shall be provided to prevent slippage when locking the handle in any chosen position. There shall be no movement of the handle when locked and under the pressure of drilling.

3.10.4 *Driving gear.* The driving gear shall be of metal, free from any imperfections, burrs, or defects of any nature which may affect its serviceability. The gear shall have smooth beveled teeth with a tooth contact on the face width of the gear teeth of not less than 3/16 inch with a minimum backlash of 0.010 inch. A rib, flange, or other means shall be provided on the outside diameter of the gear to resist the lengthwise thrust of the spindle.

3.10.5 *Crank arm.* The crank arm shall be of metal either attached or integral with the driving gear. The arm of the crank measured from the center of the driving gear to the center of the crank handle shall be not less than 1-1/2 inches nor more than 2-1/4 inches.

3.10.5.1 *Crank handle.* The crank handle shall be free to rotate on the attaching member. The outisde diameter shall have a radius or curve to form a good finger grip. The handle when attached to the driving gear shall be not less than 1-1/4 inches nor more than 2 inches in length.

3.10.6 *Driven gear (pinion).* The driven gear (pinion) shall be of metal and either pressed on or secured in a manner to be integral with the threaded steel spindle. The driven gear and chuck spindle shall be a component part of the chuck turret head when the unit is assembled.

3.10.7 *Spindle.* The spindle shall be of high-grade steel. The lower portion or chuck end of the spindle shall have a fine thread to properly accommodate the threaded hole of the chuck. The threads shall be concentric with the spindle within 0.005-inch total indicator reading throughout the entire length of the thread. The upper or smooth portion shall be machined or ground to a size in order that the maximum clearance between the spindle and bearing surface of the turret head will not exceed 0.005 inch.

3.10.8 *Chuck.* The chuck shall conform to the requirements of 3.8.3.

3.10.9 *Drill capacity.* The drill capacity of hand drills furnished shall accommodate drills from 0 up to including 0.250 inch.

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3.10.10 Type III drills shall be similar to figure 7.

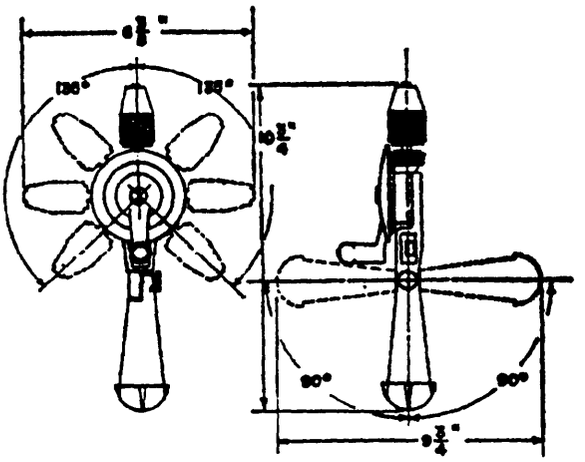


FIGURE 7.—Type III, hand, all angle adjustable hollow handle.

3.11 Type IV, push automatic. Type IV hand drills shall consist essentially of a handle, spring, actuating mechanism, chuck, and set of eight drill points. The hand drills shall have an overall length of not less than 8 nor more than 12 inches with a maximum weight of 1 pound.

3.11.1 *Handle.* The handle shall be hollow at the pressure end, capable of holding eight drill points. A removable screwcap or other means shall be provided for access to the drill magazine and be part of the hollow handle. The handle shall be well-proportioned and of a suitable shape and size to provide a comfortable handgrip.

3.11.2 The spring shall be of a material to withstand the required compression load and be capable of automatically returning the drill to its extended position.

3.11.3 *Actuating mechanism.* The actuating mechanism shall be fabricated from either bronze or other suitable metal for driving the drill points. The drive nut, spiral, and all other working parts shall be fully enclosed to prevent entry of dirt and grit.

3.11.4 *Chuck.* The chuck shall be of steel, and designed to prevent the shank of drill points from cutting or marring the gripping surfaces of the chuck. The chuck shall be designed to permit parallel movement of the gripping surface and hold concentrically the shank of drill points of a diameter within the rated capacity of the drill.

3.11.5 *Drill-point set.* The drill-point set shall consist of eight drill points ranging in diameter from 1/16 inch to 11/64 inch, inclusive, in increments of 1/64 inch and made of good quality heat-treated steel. The drill points, furnished with each drill, shall be either of the fluted-shank style (for use in a four-jawed chuck) or of the double-notch shank style (for use in a ball-locking and torque-transferring notch-type chuck) to correspond with the drill chuck furnished (see 6.2.2).

3.11.6 *Class 1, metal hollow handle.* The handle of class 1 drills shall be of metal. The drill shall be similar to figure 8.



FIGURE 8.—Type IV, class 1, metal hollow handle, push automatic.

3.11.7 *Class 2, plastic hollow handle.* The handle of class 2 drills shall be of plastic and be capable of withstanding the test specified in 4.3.4, and shall be similar to figure 9.

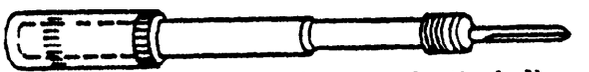


FIGURE 9.—Type IV, class 2, plastic hollow handle, push automatic.

3.11.8 *Class 3, wood hollow handle.* The handle of class 3 hand drills shall be of wood and shall be similar to figure 10.

3.12 Workmanship. Workmanship shall be first class in every respect and free from imperfections which may affect the appearance, durability and serviceability of the drills.

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FIGURE 10.—Type IV, class 3, wood hollow handle, push automatic.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling and inspection for acceptance.** Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated herein.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 *Intermediate testing.* Hardness testing specified in 4.3.5 shall be performed prior to final assembly. A sample unit shall be one jaw. For jaws heat treated as a batch and under the same process and control, the inspection level shall be S-2 with an acceptable quality level of 1.5 defects per hundred units.

4.2.3 *End product inspection.*

4.2.3.1 *Inspection of the end item.* Inspection shall be made of the end item in accordance with the following paragraphs. A lot shall consist of completely fabricated and assembled drills of the same type, class, style, and size. The sample unit shall be one drill.

4.2.3.2 *Visual and dimensional examination.* Each sample unit shall be examined for the defects as outlined in table I. Any departure from specified requirements shall constitute a defect. The inspection level shall be level II with an acceptable quality level of 4.0 for major defects and 10.0 for total defects expressed in defects per hundred units.

TABLE I.—*Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical:	None defined.
Major:	
101	Type, class, style, or capacity of drill not as specified; style of chuck not as specified.
102	Materials defective or not as specified; evidence of cracks, splits, voids, deep pits, shrinkage defects, hard spots, blowholes, or foreign inclusions.
103	Gears not machine cut, not bevel tooth style, or tooth contact less than the required minimum; evidence of fins, burrs, rough spots or improperly machined teeth.
104	Spring defective or fails to automatically return the drill to its extended position. (type IV only)
105	Actuating mechanism defective or not fully enclosed. (type IV only)
106	Outside diameter of driving gear not provided with a rib (or flange) to prevent end thrust of the spindle. (except type IV)
107	Spindle bent, not bearing mounted as required, or clearance between bearing and spindle exceeds the maximum allowable. (except type IV)
108	Spindle threads (for chuck) defective or not the specified size and fit; threads stripped, torn, cracked,

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Categories	Defects
Major:	crossed, or not concentric. (except (type IV)
109	Chuck defective, not the specified type, jaws not spring loaded, or sleeve not knurled; jaws or internal surface of sleeve not machine finished as required. (except type IV)
110	Chuck defective, fails to accommodate the specified sizes of drill points, sleeve not knurled, or (if applicable) jaws not smoothly finished. (type IV only)
111	Crank not secure on driving gear, handle not secure on crank, or handle does not revolve freely.
112	Main handle not secure on frame, wooden handle not reinforced with a ferrule, or ferrule not secure. (types I and II only)
113	Handle not secure on frame, magazine (for holding spare drills) inadequate or not readily accessible; threads on magazine cap (if applicable, crossed, or incomplete. (type III only)
114	Handle defective, fails to accommodate the required number or drill points, or magazine access (screw-cap) defective; threads stripped, torn, cracked, crossed, drunken, or not perfect full threads. (type IV only)
115	Magazine, for holding spare drills, inadequate, not readily accessible, or threads on cap or handle (if applicable) stripped, cracked, incomplete, or crossed. (type I only)
116	Handle not free to rotate in a plane perpendicular to the chuck, angle of movement less than required, or evidence of looseness or slippage when handle is locked in place. (type III only)
117	Spare drills defective or not the specified type and sizes. (type I and III only)
118	Drill point set defective, incomplete, or not the specified shank type. (type IV only)
119	Ratchet mechanism defective or does not provide the required functions. (types I and II, style A only)
120	Speed changing mechanism defective. (types I and II, class 2 only)
121	Side grip (if required) can not be securely attached to frame, not ferrule reinforced, or ferrule not se-

Categories	Defects
Major:	cure. (types I and II only)
122	Chuck turret head does not provide the required angular offset, number of positions less than required, head can not be locked in any of the positions, or gears not fully meshed in any position; evidence of sticking, binding, or excessive looseness. (type III only)
123	Bearing surfaces not smoothly finished; evidence of fins, burrs, rust, deep scratches or deep tool marks.
124	Drill not smooth operating; evidence of wobble, sticking, binding or excessive looseness; gear ratio not within the specified maximum and minimum values.
Minor:	
201	Handles not smoothly finished or not coated as required; evidence of cracking, peeling, fins, burrs, rough surfaces, sharp edges, uncoated or nonadherent areas.
202	Nonbearing surfaces not smoothly finished; evidence of rust, sand, fins, burrs, nodules, or scale.
203	Length of crank or crank handle not within the specified maximum and minimum values. (except type IV)
204	Weight of drill is less than the permissible minimum. (type III only)
205	Overall length of drill not within the specified maximum and minimum values.
206	Exposed metal surfaces not coated as specified; evidence of cracking, peeling, blisters, uncoated or nonadherent areas.
207	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.2.3.3 *Testing*. Each sample unit shall be tested as specified in 4.3 through 4.3.4.2. Any determination not within specified requirements shall constitute a defect. The inspection level shall be S-3 for 4.3.1 and 4.3.2. Inspection level S-2 shall apply to 4.3.3 through 4.3.4.2. The acceptable quality level shall be 1.5 defects per hundred units.

4.2.4 *Inspection of preparation for delivery*. Inspection to determine compliance with requirements for preparation for delivery shall be in accordance with MIL-H-15424.

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4.3 Test procedures.

4.3.1 *Concentricity test.* The frame of the drill shall be gripped in a bench vise or by other suitable means, with the chuck in a horizontal plane. A test rod of any diameter within the capacity range of the chuck to be tested, shall be inserted in the drill chuck, and the chuck or drill sleeve tightened by hand. The test rod shall project approximately 2.0 inches from the jaws when inserted to the full depth of the drill chuck. This test rod shall not run out more than .030 of an inch (average of 3 chuckings of the test rod for each sample drill tested) total indicator reading in 360 degrees rotation. The entire exposed length of the test rod shall be within these concentricity requirements.

4.3.2 *Drilling test.*

4.3.2.1 *Type I, class 1, style B, and type II, class 1, single-speed without ratchet hand drills.* Each of the drills selected shall be tested in the following manner. The drills shall be subjected to drilling a soft, cold-rolled steel sheet or plate in a vertical position. The steel sheet or plate shall be not less than 1/8 nor more than 3/16 inch thick. A maximum-size drill bit within the capacity range of the drill shall be properly inserted in the chuck jaws and the chuck tightened by hand. The crank shall be turned by hand with a forward stroke at a speed not greater than 100 revolutions per minute (r.p.m.) with a 43-pound pressure applied on large end of the handle. The longitudinal axis of the drill shall be held at approximately right angles to the steel plate. When the drill is about to pierce the opposite side of the plate, the 43-pound load shall be released and normal pressure applied on the handle until the drill has pierced entirely through the plate. Each individual drill subjected to the test shall make at least one such drilling operation to determine that the drill is in compliance with the requirements of this specification. The drill and its component members shall show no mechanical defects and shall not fail, fracture, take permanent set or show any evidence of binding or undue wear.

4.3.2.2 *Type I, class 2, style B, and type II, class 2, two-speed, without ratchet, hand drills.* The test for two-speed drills selected shall be conducted in the same manner as specified in 4.3.2.1. In addition, the drills shall be subjected to the same drilling operations in both the low and high speed. In these tests, there shall be no difficulties with the speed-changing mechanism or failure of the chucks to properly grip and hold securely the shanks of drills.

4.3.2.3 *Type I, class 1 and 2, style A, hand drill with ratchet mechanism.* The test for hand drills with ratchet mechanism selected shall be conducted in the same manner as specified in 4.3.2.1. In addition, there shall be no difficulties with the ratchet mechanism in either the low or high speed when subjected to the drilling operation.

4.3.2.4 *Type III, all angle adjustable hollow handle.* Each of the type III drills shall be tested in the same manner when in a normal position (chuck turret head and hollow handle on a longitudinal axis) as specified in 4.3.2.1. The chuck turret head of each drill shall also be adjusted and the drill subjected to further tests. At least three drilling operations shall be made in each position of the chuck turret head or test shall be of a sufficient duration to satisfy the inspector that the drill complies with the requirements of this specification. The test shall be conducted as follows:

(*a*) When the drill is held in a normal position for right-hand operation, the hollow handle shall be locked to the left in a position 90 degrees on a horizontal plane to the longitudinal axis. Also, the chuck turret head shall be swung vertically downward and to the rear 135 degrees from the longitudinal axis and locked in position and the drilling test conducted.

(*b*) In addition, the hollow handle shall remain in the 90-degree position and the chuck turret head returned

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from the 135-degree position to a position 90 degrees on a vertical plane from the longitudinal axis and locked, and a drilling test conducted.

(c) The hollow handle locked in the 90-degree position shall withstand a pressure load of 450 pounds.

4.3.2.5 *Type IV, classes* 1, 2, *and* 3, *push automatic.* Each of the drills shall be tested in the following manner: The drills shall be subjected to drilling a piece of hardwood at least 3/4 inch thick such as oak, maple, hickory, cherry, or birch in the vertical position. An 11/64-inch drill point shall be properly inserted in the chuck jaws and the chuck tightened by hand. The drill point shall not wobble or rotate eccentrically when the tool is actuated. Each drill subjected to the test shall drill at least 10 holes through the hardwood. The drill point shall not loosen or come out of the chuck during the test.

4.3.3 *Drop test for type III, sand- of die-cast hand drills.* Each representative sample of the drill shall be dropped from a height of 3 feet on a concrete floor a minimum of five times. Each sample shall then be thoroughly inspected for cracked, broken, or bent parts which are constructed from sand- or die-cast material and each drill shall be operated to check actual performance. Any drill failing to pass this test shall be rejected.

4.3.4 *Test procedures for plastic handles.*

4.3.4.1 *Heat distortion test.* The handle of the hand drills for test shall be completely immersed in boiling water at 212°F. for a period of 2 minutes. At the end of this period, the sample shall be removed from the water and shall show no significant signs of distortion or blistering.

4.3.4.2 *Flammability test.* The metal body of the drill for test shall be clamped in a support with the handle extended outward in a horizontal position. Under the handle there shall be clamped a piece of 10-mesh Bunsen-burner gauze about 5 inches square, in a horizontal position, 1/4 inch below the bottom of the handle, and with about 1/2 inch of the handle extending beyond the edge of the gauze. A Bunsen burner with an efficient flame 1/2 to 3/4 inch in height, shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds, the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handle after the removal of the Bunsen burner. In case the handle does not continue to burn after the first ignition, the burner shall be placed under the free end for a period of 30 seconds immediately following the extinction of the flame. If the handle does not continue to burn until the flame has reached the metal portion of the handle after the second ignition, the sample shall be considered self-extinguishing. When tested as specified herein, the plastic handles shall burn at a rate not exceeding 2 inches per minute.

4.3.5 *Hardness (types I, II, and III).* The chucks of the drills shall be subjected to a hardness test to determine conformance with 3.8.3, 3.9 and 3.10.8.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Level C cleaning, preservation, and packaging shall be in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with level A of MIL-H-15424.

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5.2.2 *Level B.* Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 *Level C.* Hand and push drills preserved and packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military activities.* Marking shall be in accordance with MIL-H-15424.

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The hand-operated drills covered by this specification are generally used for drilling holes in various kinds of metals, woods, plastic or other applicable materials.

6.1.1 Types I and II drills are for light, medium and heavy duty general purpose drilling.

6.1.2 Type III, all angle drills are designed for light and medium duty drilling especially in close quarters around obstructions.

6.1.3 Type IV push automatic drills are quick acting and are used primarly for light drilling.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class and style of hand drill required (see 1.2).
(*c*) Type of coating required (see 3.4.1).
(*d*) Specify metal of frame if particular ferrous metal is required for extreme ambient temperatures (see 3.8.1).
(*e*) For types I and II, drill capacity required (see 3.8.7 and 3.9).
(*f*) For drill-point sets for type IV, procured separately, specify shank (see 3.11.5 and 6.2.2).
(*g*) Levels of preservation and packaging and packing required (see 5.1 and 5.2).

6.2.1 When repair parts are required, the contract or order should specify the following:

(*a*) The manufacturer's name or trademark of the drills for which the part is required.
(*b*) Proper name of the part required.
(*c*) Quantity of each item required.
(*d*) Other data or description applicable to the hand drills which will assist in identifying the proper repair parts.

6.2.2 *When drill-point sets are procured separately from the push automatic drills, type IV.* When drill-point sets for push automatic drills, type IV, are procured separately from the drills, the procuring agency shall specify either fluted-shank style or double-notch shank style as required (see 3.11.5).

6.3 Superseding data. Type I, class 2, style A of Federal Specification GGG-D-671b has been discontinued by industry and has been deleted from this specification.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Drills, hand, not otherwise indexed by name
Carload minimum weight 80,000 pounds.

Motor:
Tool, hand, not otherwise indexed.

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Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—Ships

Air Force—MOAMA

Preparing Activity:

Navy—Ships

Copies of this specification may be purchased for 10 cents each.

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